

# AJEER MOHAMMED

Senior Full-Stack Engineer | Web, Mobile and AI Systems

India | Open to relocation and remote roles

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## SUMMARY

Full-stack engineer building production web, mobile and AI systems, and running the infrastructure behind them. Currently the lead developer on a multi-tenant e-commerce platform and an AI shopping assistant, where the work covers architecture, planning, cost decisions and code review for a team of 4 to 6 developers. Past projects have shipped to a US state health department, international pharmaceutical brands, and retail businesses in India, the UAE and the United States. Also publish open-source developer tooling: two command-line tools for catching expensive mistakes in a build rather than in production.

**Selected work:** made tenant isolation a build-time guarantee on a 56-model multi-tenant platform, so a query that could leak one merchant's data into another store fails CI instead of reaching customers.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript (ES6+), Go, Dart, Python, PHP, SQL

**Frontend:** React 19, Next.js (App Router), React Router 7, Angular, Flutter, Tailwind CSS, Redux, Riverpod, TanStack Query

**Backend and Data:** Node.js, NestJS, Fastify, Express.js, Prisma, PostgreSQL, Supabase, Redis, MongoDB, BullMQ, REST APIs

**AI and Agentic Engineering:** LLM integration (Claude, Gemini, Groq/Llama), RAG, vector search and embeddings, tool-calling agents, SSE streaming, provider fallback, agent instructions and CI safety gates for codebases, test and verification harnesses

**E-commerce:** Shopify Hydrogen and Oxygen, Shopify App Remix, Polaris, App Bridge, Sanity CMS, Razorpay

**Infrastructure and Delivery:** Docker, GitHub Actions CI/CD, Cloudflare Workers, Vercel, Railway, GCP, Firebase, Nginx, PM2, pnpm and Turborepo monorepos, Jest, Vitest, Cypress, Playwright, Git

**Ways of Working:** Multi-tenant architecture, technical planning and decision records, cost modelling, security review, code review and mentoring, Agile delivery

## PROFESSIONAL EXPERIENCE

**SEA Media** | Senior Full-Stack Engineer and Project Partner

India (Remote) | 05/2025 - Present

- Enrixa Store:** built a multi-tenant e-commerce platform, now 56 data models and 259 REST endpoints across 48 controllers, where every merchant runs an isolated store on its own subdomain. NestJS, Prisma, PostgreSQL and Redis on the backend, with separate Next.js apps for the merchant admin and the customer storefront. Set the tenancy model, the payments layer covering Razorpay and cash on delivery, and the release plan.
- Closed the platform's biggest security risk by turning tenant isolation into a build requirement. Every pull request runs an automated isolation check and a migration compatibility diff against main across a schema that has taken 83 migrations, so a query that could leak one merchant's data into another store fails CI rather than reaching customers.
- Enrixa AI:** shipped an AI shopping assistant for Shopify. A Fastify service streams tool-calling agent replies over SSE, BullMQ workers keep the product catalog and its embeddings in sync, and the storefront chat widget runs inside a shadow DOM so it never collides with theme CSS. Wrote the product and technical plan behind it, including the cost model and the scaling approach, and kept a decision log for choices that would be expensive to reverse.
- W.I.N.S, Alabama Department of Public Health:** led a two-repository build for an infant safety program. A Next.js 16 and React 19 admin CMS on Supabase, plus a Flutter app for parents tracking their child's health, milestones and vaccinations, both sharing one PostgreSQL database of 15 tables across 11 edge functions. Wrote the schema contract and table ownership rules that let two teams work on the same database without breaking each other's features, backed by about 100 test files across the two repositories. Live on the App Store and Google Play since July 2026 as the state agency's official app for the program.
- Kidulan:** built a Shopify Hydrogen storefront serving more than 10,000 daily users across India, 96 routes on React Router 7, Oxygen and Sanity CMS with four other engineers, and owned the architecture notes and the Oxygen deploy pipeline.
- Price Monitor:** a Shopify app that pulls in competitor catalogs, matches their products against the merchant's own, and adjusts prices by rule. Repricing stays behind a dry run until the merchant approves it.
- Wrote the agent instructions and CI gates that let AI coding agents work in 8 production repositories, including automated audit, tenant-check and review-watcher scripts. Agents pass through the same gates the team does, so their changes still get reviewed and verified before they land. The generalised version of those gates is published as **bouncer**, below.
- Handled deployment and day-to-day operations across all of it: Docker images, GitHub Actions pipelines, Vercel and Shopify Oxygen releases, PM2 process management, and the managed PostgreSQL and Redis instances. Keep roughly 185 test files green across the portfolio.

## NTAM Group (Delta Plus Event Management) | Software Developer

Dubai, UAE | 06/2024 - 04/2025

- Built a gamified training platform for Abbott's medical representatives, used by over 1,000 reps worldwide, with per-region admin panels, a CMS and reporting. Angular, PHP, Node.js, SQL and microservices.
- Delivered L'Oréal's event app for Android and iOS, covering agendas, badge scanning, surveys, live voting and push notifications. Used at more than 5 live events. Flutter, .NET and SQL.
- Built the interactive pieces for Nestlé's Ascenda launch at the Ritz-Carlton, Jeddah: a full-body-tracking growth wall where guests watched themselves grow on a 4 metre screen, then printed or shared the photo, and a Mario-style runner played on an Xbox controller. More than 150 guests used them in one evening. Electron, React, Redux and MediaPipe.
- Built a talking avatar chatbot for Novartis that answered product questions for healthcare professionals, using Next.js, Python, Google TTS and a fine-tuned model.

## Indus Real Estate LLC | Full-Stack Developer

Dubai, UAE | 12/2022 - 06/2024

- Built a lease management CRM as a progressive web app covering more than 200 units, tracking lease expiry, payments, cheques, maintenance and reporting. Angular, PHP, Node.js and SQL on Linux.
- Rebuilt the public property site with better filtering and agent contact flows, cutting page load time by 70 percent by fixing first contentful paint. Next.js, Node.js, PHP and SQL.
- Added a CMS with an image pipeline that resizes, crops and converts uploads to WebP through Sharp, along with reporting tools for the content team.

## DOSII | Flutter Developer

India | 06/2022 - 12/2022

- Helped build a university admissions platform with predictive analytics that served over 1,000 students in its first quarter, and set up the project's unit and integration tests. Flutter, Python, Firebase, GCP and GetX.

## OPEN SOURCE

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**bouncer** - checks every code change for five expensive mistakes before they merge, in CI and inside the editor next to an AI agent. MIT, on npm as **bouncer-gates**.

Five checks: leaked secrets, database queries that cross tenants, float arithmetic on currency, migrations that break the running app during a deploy, and documentation links pointing at deleted files. Each gate is a pure function taking files and returning findings, so one implementation runs in CI as a GitHub Action, in Claude Code and Cursor as an MCP server and post-edit hook, in a Cloudflare Worker behind the live playground, as a Node service on Railway, and in the browser as a fallback. Zero runtime dependencies, 115 tests.

Most of the work went into being wrong less often: an early build reported 45 blocking findings on a 1,209-file repository, of which 32 were false positives. Reading all 45 by hand and fixing the causes brought it to 13 findings, all genuine. Every one of the 32 is now a test asserting the gate stays quiet.

[github.com/ajeermahmood/bouncer](https://github.com/ajeermahmood/bouncer) | [npmjs.com/package/bouncer-gates](https://npmjs.com/package/bouncer-gates)

**estate** - a linter for repositories rather than for the code inside them. MIT, one Go binary.

Checks a directory of repositories in parallel against declarative rules and exits non-zero, so it behaves the same in a terminal and in CI: whether a repository has a remote configured at all, and whether a **.env** would be ignored by the next commit, among others. The first real run found a repository holding six months of work that existed on one disk with nowhere to push. Zero third-party dependencies, 102 tests, cross-compiled for Linux, macOS and Windows on x86-64 and ARM64.

[github.com/ajeermahmood/estate](https://github.com/ajeermahmood/estate)

**Two small npm utilities:** web-element-scraper (extracts text from page elements with Puppeteer and Cheerio) and nodejs-performance-profiler (reports memory, CPU and uptime for Node.js applications).

## WRITING

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**Making a codebase safe for AI coding agents** - on why instructions files are not enough to let coding agents work in a production repository, and which rules belong in a linter, a compiler, an instructions file, or a blocking CI gate. [ajeer.website/blog](https://ajeer.website/blog)

## EDUCATION

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**Crossroads Software Engineering Bootcamp** | India | 08/2020 - 11/2021

Mobile and web development with backend services. Delivered more than 10 projects in agile teams.